

Asthma — Pharmacology Pharmacy (Theophylline, AERD, ICS side effects)



1. Theophylline narrow window

WHAT YOU SEE

A theophylline bottle on a tightrope 'narrow therapeutic window — rarely used today'.

WHAT IT ENCODES

Theophylline is a methylxanthine that inhibits phosphodiesterase (raising cAMP) and antagonizes adenosine receptors, giving bronchodilation plus mild anti-inflammatory effect. Its therapeutic range is narrow (~10–20 µg/mL), so it requires serum-level monitoring and is now relegated to add-on therapy when biologics/LAMA are unavailable. It is metabolized by CYP1A2.

BOARD TRAP

Theophylline is NOT first-line in modern guidelines — toxicity (arrhythmia, seizures) appears at levels only modestly above therapeutic, and many factors (heart failure, cirrhosis, fever, age, CYP1A2 inhibitors) shift levels into the toxic range.

2. PDE inhibitor → cAMP → bronchodilation

WHAT YOU SEE

A muscle figure 'cAMP ↑ → bronchodilation' from PDE inhibition.

WHAT IT ENCODES

Phosphodiesterase normally degrades cAMP; inhibiting it raises intracellular cAMP in airway smooth muscle, causing relaxation and bronchodilation. This is the same downstream second messenger that beta-2 agonists raise by stimulating Gs-coupled receptors — the two converge on cAMP. Roflumilast (PDE4 inhibitor) uses the same principle in COPD.

3. Theophylline toxicity (tachycardia/seizures)

WHAT YOU SEE

An EKG with 'tachycardia, seizures, ineffective' near a lightning bolt.

WHAT IT ENCODES

Theophylline overdose produces the triad of tachyarrhythmias (sinus tach, SVT, atrial/ventricular arrhythmias), CNS excitation (seizures, which are often refractory), and GI upset (nausea, vomiting). Severe toxicity also causes hypokalemia, hypophosphatemia, hyperglycemia, and metabolic/lactic acidosis. Treatment is supportive plus multiple-dose activated charcoal; hemodialysis for severe levels.

BOARD TRAP

CYP1A2 inhibitors (ciprofloxacin, fluvoxamine, cimetidine, erythromycin) raise theophylline levels into the toxic range — always check interactions before adding any new drug, and seizures in theophylline toxicity may need benzodiazepines/barbiturates (avoid phenytoin).

4. Osteoporosis (high-dose ICS)

WHAT YOU SEE

A bone labeled 'osteoporosis — high dose' with cataracts/glaucoma/growth-suppression notes.

WHAT IT ENCODES

High-dose or long-term inhaled corticosteroids carry dose-dependent systemic effects: reduced bone mineral density/osteoporosis, posterior subcapsular cataracts, glaucoma, easy bruising/skin thinning, and a small linear-growth reduction in children (~0.5 cm in the first year, largely catch-up). Adrenal suppression is possible at very high doses.

BOARD TRAP

ICS are far safer than oral steroids but are NOT systemically inert at high doses — use the lowest effective dose, and the catch-all 'inhaled steroids have no systemic effects' is a false board statement.

5. ICS oral thrush (candida)

WHAT YOU SEE

An ICS inhaler with 'candida — oral thrush; use spacer + gargle'.

WHAT IT ENCODES

The most common LOCAL adverse effects of ICS are oropharyngeal candidiasis (thrush) and dysphonia/hoarseness from laryngeal myopathy. Both are prevented by using a spacer and rinsing/gargling the mouth after each dose. Local immunosuppression of the oral mucosa allows Candida overgrowth.

BOARD TRAP

ICS cause Candida thrush — NOT HSV (the herpes option is crossed out in the scene). The classic distractor pairs ICS with a viral local infection; the correct association is fungal candidiasis plus dysphonia.

6. ASA / COX-1 (AERD)

WHAT YOU SEE

A figure 'ASA' inhibiting 'COX-1' with arrows to leukotrienes.

WHAT IT ENCODES

Aspirin and nonselective NSAIDs irreversibly/competitively block COX-1, removing the bronchoprotective prostaglandin E2 (PGE2) brake and shunting arachidonic acid down the 5-lipoxygenase pathway to overproduce cysteinyl leukotrienes (LTC4/D4/E4). In AERD patients this causes acute bronchospasm, nasal congestion, and flushing within 30–180 minutes of NSAID exposure.

BOARD TRAP

It is COX-1 inhibition (not COX-2) that triggers AERD reactions — therefore COX-2-selective agents (celecoxib, etoricoxib) and acetaminophen are generally tolerated, while any nonselective NSAID (ibuprofen, naproxen, ketorolac) can precipitate a severe reaction.

7. AERD triad (5-10% of asthmatics)

WHAT YOU SEE

A panel 'AERD — 5-10% of asthmatics' with severe asthma, nasal polyposis, eosinophilic sinusitis.

WHAT IT ENCODES

Aspirin-exacerbated respiratory disease (Samter's triad) = severe persistent asthma + recurrent nasal polyposis/eosinophilic chronic rhinosinusitis + NSAID hypersensitivity, affecting ~5–10% of adult asthmatics (higher among severe asthmatics). It is a non-IgE-mediated, leukotriene-overproduction disorder. Management: leukotriene modifiers, ICS, and aspirin desensitization for refractory disease; dupilumab is highly effective for the polyps and asthma.

BOARD TRAP

The classic vignette — adult with severe asthma, recurrent nasal polyps despite surgery, and a wheezing/flushing reaction to ibuprofen = AERD. Treat with LTRA/zileuton and consider aspirin desensitization; do NOT simply re-challenge with NSAIDs.

8. COX-2 / paracetamol safe

WHAT YOU SEE

Bottles 'celecoxib, etoricoxib, paracetamol' marked safer in AERD.

WHAT IT ENCODES

Because AERD is driven by loss of COX-1-derived PGE2, COX-2-selective inhibitors (celecoxib, etoricoxib) and low-to-moderate-dose acetaminophen/paracetamol are tolerated by the large majority of AERD patients and are the preferred analgesics/antipyretics. High-dose acetaminophen (>1 g) can rarely cross-react.

BOARD TRAP

When an AERD patient needs an analgesic, choose a COX-2-selective agent or paracetamol — selecting any nonselective NSAID is the wrong board answer.

9. Zileuton (5-LOX inhibitor, LFT risk)

WHAT YOU SEE

A 'ZILEUTON 5-LOX inhibitor' bottle with 'LFTs in 3%, CYP1A2'.

WHAT IT ENCODES

Zileuton inhibits 5-lipoxygenase (5-LOX), the enzyme upstream of all leukotrienes, blocking both bronchoconstrictor cys-LTs and chemotactic LTB4 — broader than montelukast/zafirlukast which only block the CysLT1 receptor. It is dosed multiple times daily and is particularly useful in AERD.

BOARD TRAP

Zileuton causes hepatotoxicity (ALT elevation in ~2–4%) and requires baseline and periodic LFT monitoring; it inhibits CYP1A2 and therefore RAISES theophylline and warfarin levels — a double board trap linking it back to theophylline toxicity.

10. Montelukast neuropsychiatric warning

WHAT YOU SEE

A dark cabinet 'MONTELUKAST — neuropsychiatric black box warning: depression/suicidal ideation'.

WHAT IT ENCODES

Montelukast is a CysLT1-receptor antagonist (LTRA) used for mild asthma, EIB, allergic rhinitis, and AERD. In 2020 the FDA added a BOXED WARNING for serious neuropsychiatric events — agitation, depression, sleep disturbance, and suicidal thoughts/behavior — recommending it be reserved when other therapies are inadequate.

BOARD TRAP

Montelukast is not benign — counsel on mood/behavior changes and reconsider it for allergic rhinitis if safer options exist. LTRAs are also generally less effective than ICS as controllers and should not replace ICS in persistent asthma.

11. ACE inhibitor board trap

WHAT YOU SEE

A 'board traps' box: ACE inhibitor with a red X.

WHAT IT ENCODES

ACE inhibitors block bradykinin breakdown; accumulated bradykinin and substance P cause a dry, non-productive cough in ~5–20% of users, which can be mistaken for or aggravate asthma symptoms. The cough is class-specific (not dose-related) and resolves days to weeks after stopping.

BOARD TRAP

A new dry cough after starting an ACE inhibitor is the DRUG, not worsening asthma — switch to an ARB (which does not raise bradykinin). Don't escalate asthma controllers in response.

12. Beta-blocker board trap

WHAT YOU SEE

A 'beta-blocker' card with a red X next to statin.

WHAT IT ENCODES

Nonselective beta-blockers (propranolol, timolol — including ophthalmic drops) block bronchial beta-2 receptors and can precipitate bronchospasm in asthmatics. When a beta-blocker is genuinely indicated (e.g., post-MI), a cardioselective beta-1 agent (metoprolol, bisoprolol, atenolol) at low dose is preferred and usually tolerated.

BOARD TRAP

Avoid NONselective beta-blockers in asthma; cardioselective ones may be used cautiously when essential. Statins are SAFE and crossed out in the scene as a non-trigger — don't pick statin as a respiratory offender.